

GE Healthcare

MR Quiet Upgrade



OPERATING DOCUMENTATION

5670077
Revision 3.0



WARNING



STRONG MAGNETIC FIELD



NO PACEMAKERS*
NO NEUROSTIMULATORS*
NO CONDUCTIVE/METALLIC IMPLANTS*



**Persons with pacemakers,
neurostimulators or metallic
implants must not enter this area.**
Serious injury may result.

* In general, patients with conductive (e.g. metallic) implants are contraindicated for MR scans. For patients with implants that are labeled as 'MR Safe' or 'MR Conditional', consult the implant device manufacturer's documentation.

* WARNING: Only use quadrature transmit for 'MR Conditional' devices.

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Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. - Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. - Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. - Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 - 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 - 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 - 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 - 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 - 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 - 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

**VÝSTRAHA
(CS)**

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

**ADVARSEL
(DA)**

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

**WAARSCHUWING
(NL)**

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

**WARNING
(EN)**

This service manual is available in English only.

- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

**HOIATUS
(ET)**

See teenindusjuhend on saadaval ainult inglise keeles.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

**VAROITUS
(FI)**

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

**ATTENTION
(FR)**

Ce manuel d'installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

**WARNUNG
(DE)**

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendienst-technikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

**ΠΡΟΕΙΔΟΠΟΙΗΣΗ
(EL)**

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

**FIGYELMEZTETÉS
(HU)**

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

**ÁÐVÖRUN
(IS)**

Þessi þjónustuhandbók er aðeins fánleg á ensku.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

**AVVERTENZA
(IT)**

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

**警告
(JA)**

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

**경고
(KO)**

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

**BRĪDINĀJUMS
(LV)**

Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

**ĮSPĖJIMAS
(LT)**

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

**ADVARSEL
(NO)**

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

**OSTRZEŻENIE
(PL)**

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

**ATENÇÃO
(PT-BR)**

Este manual de assistência técnica encontra-se disponível unicamente em inglês.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

**ATENÇÃO
(PT-PT)**

Este manual de assistência técnica só se encontra disponível em inglês.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

**ATENȚIE
(RO)**

Acest manual de service este disponibil doar în limba engleză.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

**ОСТОРОЖНО!
(RU)**

Данное руководство по техническому обслуживанию представлено только на английском языке.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

**UPOZORENJE
(SR)**

Ovo servisno uputstvo je dostupno samo na engleskom jeziku.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodi-lačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

**UPOZORNENIE
(SK)**

Tento návod na obsluhu je k dispozícii len v angličtine.

- Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

**ATENCION
(ES)**

Este manual de servicio sólo existe en inglés.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

**VARNING
(SV)**

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

**OPOZORILO
(SL)**

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

**DİKKAT
(TR)**

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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Revision History

Revision	Date	Description
1.0	May 1, 2013	Initial release.
2.0	July 23, 2013	Updated revision to assign part classification – no content change
3.0	July 23, 2013	Updated revision to synchronize revision history with MyWorkshop – no content change

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Table of Contents

Chapter 1 OVERVIEW	15
1 Getting Started	15
1.1 Upgrade Introduction	15
1.2 Site Ready Check	15
1.3 Pre-Delivery FDO Checks	15
1.4 Upgrade Schedule Considerations	15
1.5 Product Delivery Instructions and Packing	15
1.6 Damage in Transportation	16
1.7 Product Locator	16
1.8 Required Equipment and Tools	17
1.9 Discarded Material Return Policy	17
1.9.1 Material Policy Overview	18
1.9.1.1 Returned Material	18
1.9.1.2 Disposed Material	18
1.9.2 Material to be Returned	18
1.9.3 Return Kit	18
1.9.4 Deinstallation Procedures	18
1.9.5 Region Specific Return Information	18
1.9.5.1 Sites in Americas	18
1.9.5.2 Sites in Asia	19
1.9.5.3 Sites in Europe	19
1.9.6 Product Locator Information	19
1.10 Upgrade Components	19
1.10.1 Components Supplied for MR450w GEM Systems (M7001SB)	20
1.10.2 Components Supplied for MR750w GEM Systems (M7001SD)	21
Chapter 2 UPGRADE PROCEDURE	23
1 Upgrade Procedures	23
1.1 Pre-Mechanical Upgrade Procedures	23
1.2 Mechanical Upgrade Procedures	23

1.2.1 Power Down CAM Chassis.....	23
1.2.2 Board Replacement Identification.....	25
1.2.2.1 MR450w Board Positions.....	25
1.2.2.2 MR750w Board Positions.....	26
1.2.3 Circuit Board Removal.....	26
1.2.4 Circuit Board Replacement.....	27
1.2.5 Return Removed Modules.....	28
1.3 Post-Mechanical Upgrade Procedures.....	28
1.4 Completion.....	30
Chapter 3 APPENDIX.....	31
1 Upgrade Test Record.....	31

Chapter 1 Overview

1 Getting Started

1.1 Upgrade Introduction

This publication documents the mechanical installation information for upgrading a MR450w GEM or MR750w GEM to MR Quiet.



WARNING

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1.2 Site Ready Check

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

1.3 Pre-Delivery FDO Checks

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1.4 Upgrade Schedule Considerations

FMI, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1.5 Product Delivery Instructions and Packing

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Upgrade Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1.6 Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period. File a report with:

- Call 1-800-548-3366. Select "Install Support Services for FOA and MIS"
- Contact your local service coordinator for more information on this process.

1.7 Product Locator

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the FE Site Verification Web Site at:

<http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page - <http://supportcentral.ge.com/15563>

2. One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1.8 Required Equipment and Tools



NOTICE

Follow all required safety and PPE procedures customary for your organization when working on this product.

Table 1-1: General Safety Requirements

Item	Description
1	Wear composite safety shoes for foot protection.
2	Wear gloves for hand protection.
3	Wear safety glasses for eye protection.



DANGER

FERROUS MATERIAL HAZARD!

IF MAGNET IS ENERGIZED, TOOLS, COMPONENTS, AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

Tools to upgrade the system are listed in following table:

Table 1-2: Installation Equipment

Item	GE Part Number	Description
1		Screwdriver, slotted (long & narrow)
2	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
3		Service Methods DVD
4	46-198094P1	Wrist grounding strap
5	---	Plastic or aluminum flashlight
6	5307511-3 or (5434817-20 & 5434817 set)	RF Power Measurement Kit
7	46-194427P284	Digital Multi-meter (DMM)

1.9 Discarded Material Return Policy

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1.9.1 Material Policy Overview

1.9.1.1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly to Renewable Resources. Do not remove or swap any parts from the material to be returned.

1.9.1.2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1.9.2 Material to be Returned

All deinstalled UPM processor boards must be returned to GE Renewable Resources for recycling. If a 3.0T system, also return the GEM HNU coil.

- Mark each box: MR QUIET UPGRADE
- Send the boxes to the following address:

GE Renewable Resources
Attn: Asset Recovery Leader, Harvest
7624 South 10th Street
Oak Creek, WI 53154

1.9.3 Return Kit

If needed, packaging and/or transportation details can be obtained through the local Installation Specialist.

Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.

1.9.4 Deinstallation Procedures

Refer to instructions in this manual for deinstallation procedures of site.

1.9.5 Region Specific Return Information

1.9.5.1 Sites in Americas

All deinstalled UPM processor boards must be returned to GE Renewable Resources for recycling. If a 3.0T system, also return the GEM HNU coil.

- Mark each box: MR QUIET UPGRADE
- Send the boxes to the following address:

GE Renewable Resources
Attn: Asset Recovery Leader, Harvest
7624 South 10th Street
Oak Creek, WI 53154

1.9.5.2 Sites in Asia

Return procedure for GEMS–Asia countries, contact:

GE HEALTHCARE JAPAN CORPORATION
ATTN: Yoshiaki Tsujii (SWaP/HARVEST)
67-4, Takakura-machi, Hachioji-shi-Tokyo
192-0033 Japan
Contact at Destination: 81-426-48-2595
FAX: 81-426-48-2902

1.9.5.3 Sites in Europe

For non-Gold Seal returns in Europe:

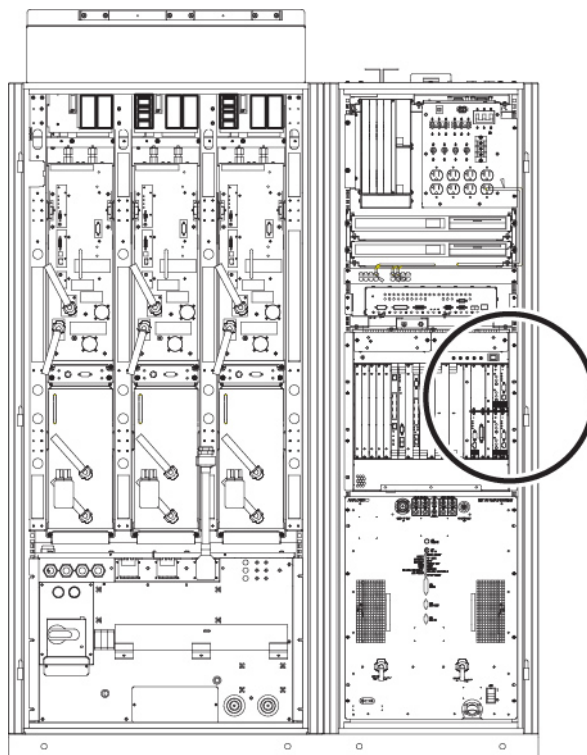
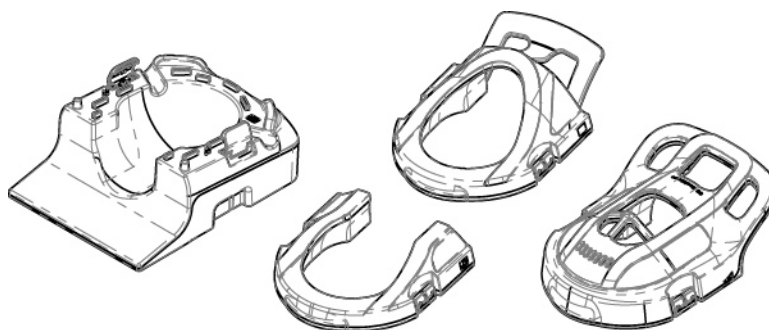
European Renewable Resources Center
GEODIS LOGISTICS - CD 26 - Pièce de la Remise - Bâtiment EVL2
Quai 45 à 49 - 91090 LISSES - FRANCE
TEL = (+33)-1-64-97-55-41 FAX = (+33)-1-60-86-12-31

1.9.6 Product Locator Information

Process product locator information for all serialized components removed from the system according to policy. Refer to [Section 1.7](#).

1.10 Upgrade Components

Dependent on system being upgraded, the following components are provided for upgrading a MR450w GEM or MR750w GEM sites.

Illustration 1-1: Location of Replacement Boards in PGR Cabinet**Illustration 1-2: 3.0T Quiet GEM HNU Coil**

1.10.1 Components Supplied for MR450w GEM Systems (M7001SB)

- Quantity 2 of RF Detector boards
- Quantity 2 of UPM Processor boards
- DV24 software collector
- Silent MR option key
- MR Quiet Upgrade manual

1.10.2 Components Supplied for MR750w GEMSystems (M7001SD)

- Quantity 4 of RF Detector boards
- Quantity 2 of UPM Processor boards
- DV24 software collector
- Silent MR option key
- 3.0T Quiet GEM HNU coil
- MR Quiet Upgrade manual

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Chapter 2 Upgrade Procedure

1 Upgrade Procedures

The general sequence of this upgrade procedure is:

[Section 1.1](#) Pre-Mechanical Upgrade Procedures

[Section 1.2](#) Mechanical Upgrade Procedures

[Section 1.3](#) Post-Mechanical Upgrade Procedures

[Section 1.4](#) Completion

1.1 Pre-Mechanical Upgrade Procedures

Refer to the applicable MR Service Methods DVD to complete the following procedures.

1. Verify that all required upgrades have arrived complete and site is ready to shut down.
2. Do check scan to verify proper system operation – see service methods Check Scan (Good-Bye)
3. To confirm the UPM Processor boards are ready to harvest, run the following two UPM diagnostics. If the diagnostic completes, the UPM Processor boards are good to return. Fill in the attached Upgrade Test Procedure page located in the **Chapter 3, Appendix** of this manual. Make sure the test procedure is dated, signed, and all the * entries are completed.
 - Analog Data
 - I/O Data

NOTE: The diagnostic location is:

- Diagnostics >> System Function >> RF >> UPM Diagnostics

OR

- Diagnostics >> Hardware Location >> PGR Cabinet >> CAM >> UPM Diagnostics

4. Perform Save Info process – see service methods Save and Restore Tool

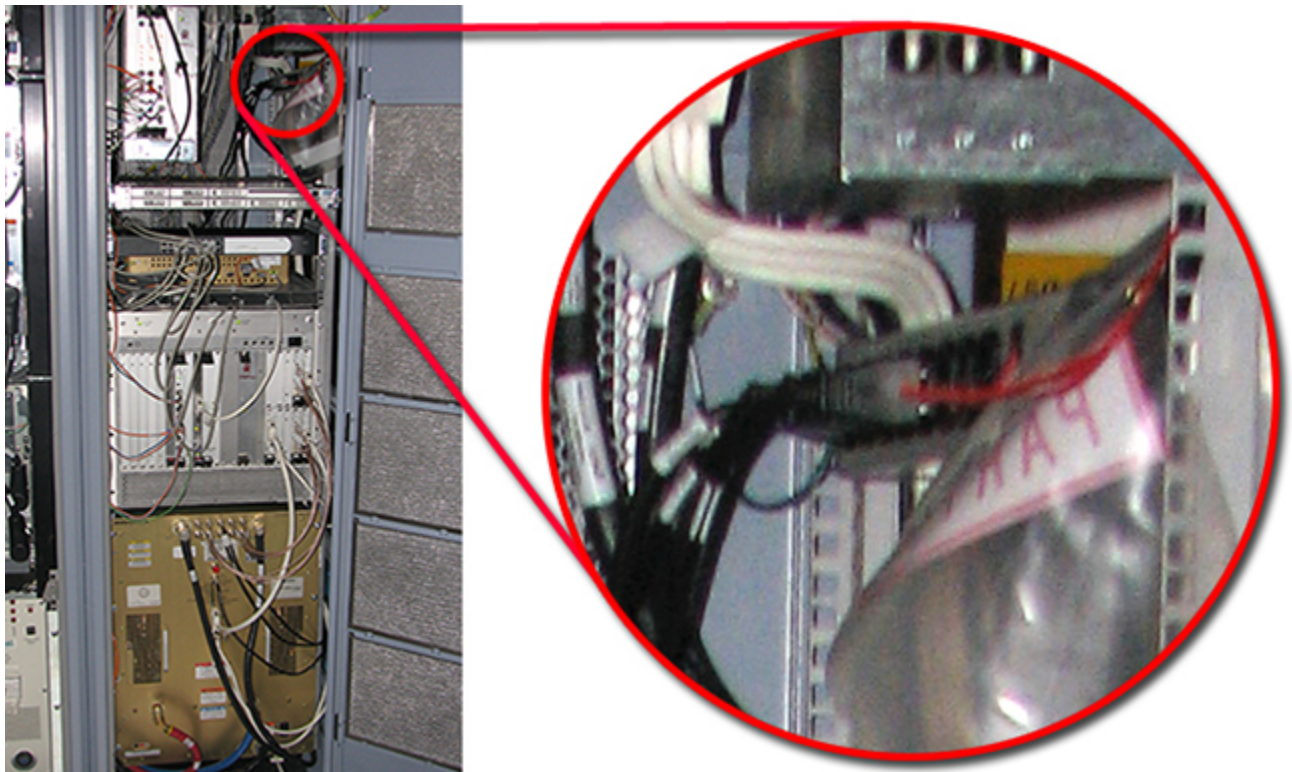
1.2 Mechanical Upgrade Procedures

1.2.1 Power Down CAM Chassis

Prior to powering down the entire CAM Chassis, power down the ICNs from the VRE power control. The VRE System, which resides in the PGR Cabinet, can become corrupted if it loses power when the applications software is active.

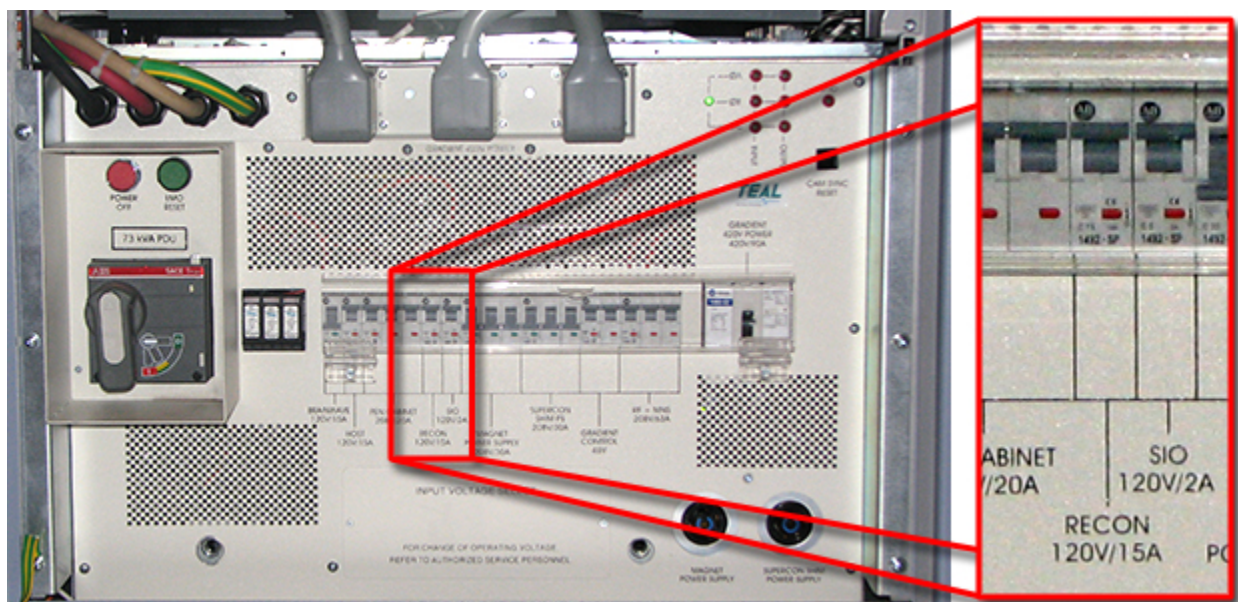
1. Use DMM to check unused outlet on terminal strip for presence of 115 VAC.

Illustration 2-1: Service Power Strip



2. At the host computer, open the service browser and select **Utilities > Tool Box > VRE > Power Control**
3. Next, select [Power Off VRE]
4. At the front panel of the PDU, switch the RECON circuit breaker to the OFF (down) position.

Illustration 2-2: RECON Circuit Breaker on PDU



5. At the front of the PDU, switch the PEN Cabinet or PDM-RF breaker to the OFF (down) position.
6. At the front panel of the PDU covering the circuit breakers, apply a personal LOTO device for each FE working on the equipment.

Illustration 2-3: PDU LOTO Device

7. Confirm that the power is OFF. Check the service power strip for 115 VAC power with the DMM. If the power is still ON, confirm that the RECON CB on the PDU is in the OFF position.

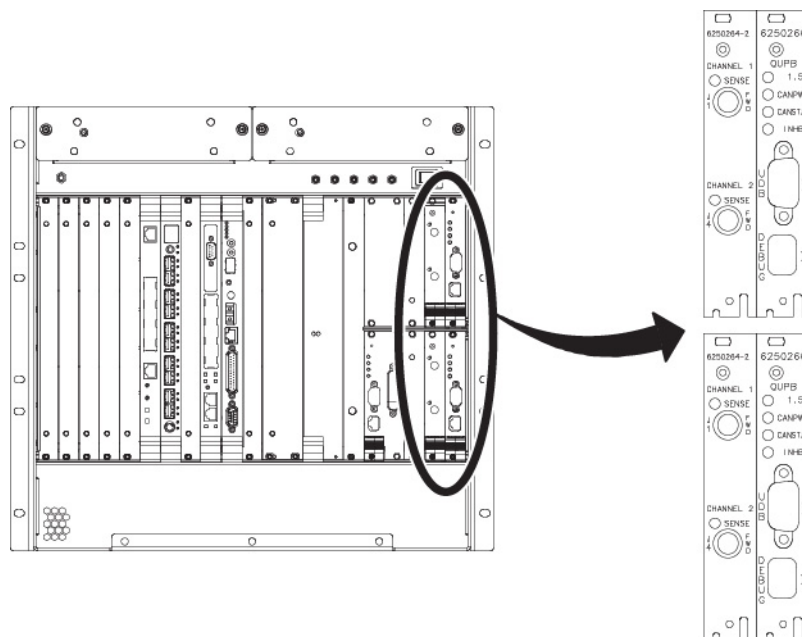
1.2.2 Board Replacement Identification

Refer to one of the following two sections, dependent on system being upgraded (MR450w or MR750w), for the applicable board configuration,

1.2.2.1 MR450w Board Positions

For slots 20U and 20L, two new RF Detector boards (6250264-2) are to be installed.

For slots 21U and 21L, two new UPM Processor boards (6250266) are to be installed.

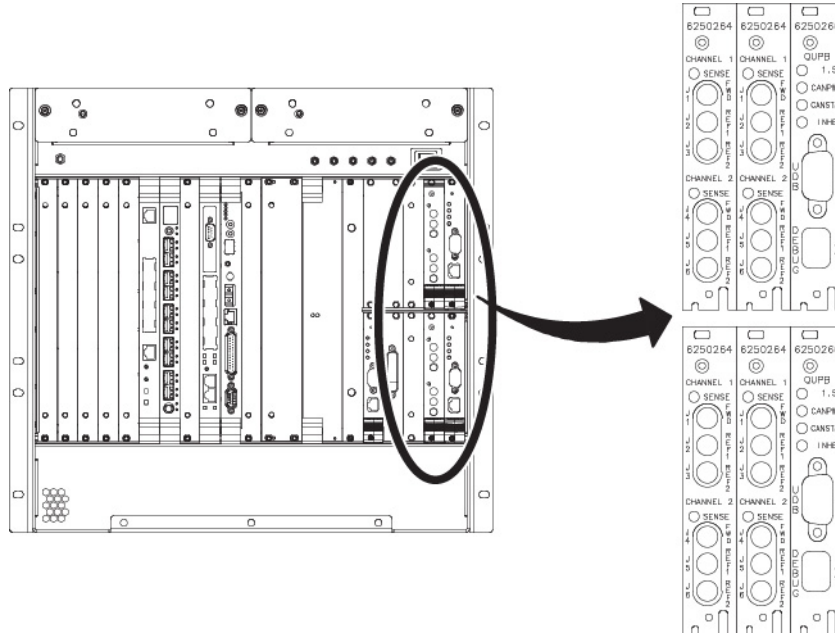
Illustration 2-4: MR450w GEM Boards

1.2.2.2 MR750w Board Positions

For slots 19U, 19L, 20U and 20L, four new RF Detector boards (6250264) are to be installed.

For slots 21U and 21L, two new UPM Processor boards (6250266) are to be installed.

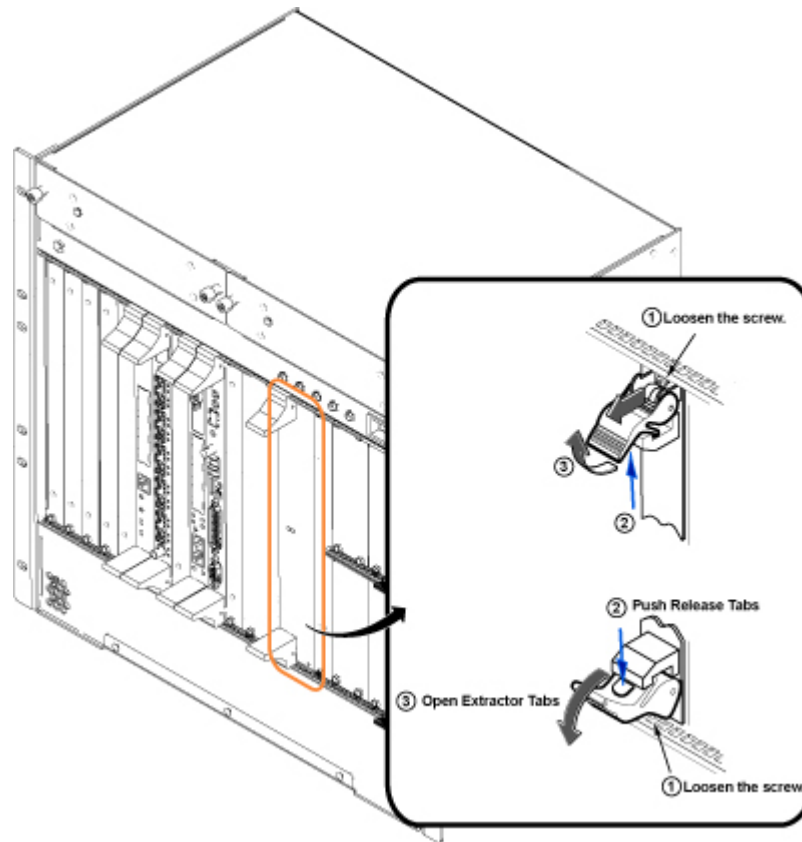
Illustration 2-5: MR750w GEM Boards



1.2.3 Circuit Board Removal

1. Attach an ESD grounding wrist strap.
2. Where applicable, remove the cables (Sub D, SMB, etc.) from the face of the board.
NOTE: When removing the cables, verify that the cable labels are accurate.
3. Loosen the faceplate holding screw. (These screws are captive to the faceplate, and do not need to be completely removed.)
4. Using your thumb, push in the gray release tab and gently pull the extractor tab located on the bottom of the circuit board. The lever action of the ejector disengages the circuit board connections from the chassis.

Illustration 2-6: Board Extractor Tabs



5. Being careful not to bend or bind the card in any way, slide the circuit board all the way out of the card cage.

NOTE: Handle the circuit boards using only the faceplate, ejectors, and the edges of the board.

6. Place the UPM Processor boards, with the completed test procedure, in an appropriate shipping package to return for recycling. The Detector boards should be disposed of per local regulations. Refer to the *Disposed Material* section in [Chapter 1, Overview](#).

1.2.4 Circuit Board Replacement

1. Remove the packing material from the applicable replacement board. Position board in assigned slot and gently slide the circuit board along the slot guides into the chassis.
2. Gently engage the circuit board connectors with the backplane.
 - a. If the circuit board is not properly engaging the backplane, remove the board and check the connectors for bent or broken pins.
 - b. Once the board ejectors contact the CAM chassis faceplate, gently push the ejector into the center of the board. This causes the ejector to insert the board into the chassis. (Do **not** attempt to use anything other than the ejector to insert or remove the board from the chassis.)

3. Tighten, but do not overtighten, the faceplate holding screws to secure the circuit board to the card rack.
4. Reattach the cables to the face of the board.

1.2.5 Return Removed Modules

All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy. Refer to [Chapter 1, Discarded Material Return Policy](#)

1.3 Post-Mechanical Upgrade Procedures

1. Remove LOTO. Restore power.
2. Load DV24 software — see service methods Loading Host System Software.
 - When in Guided Install the second time (when restoring all SaveInfo files), go to the Hardware Config menu and select “QUPM 1 Mhz Sampling Rate” for UPM type

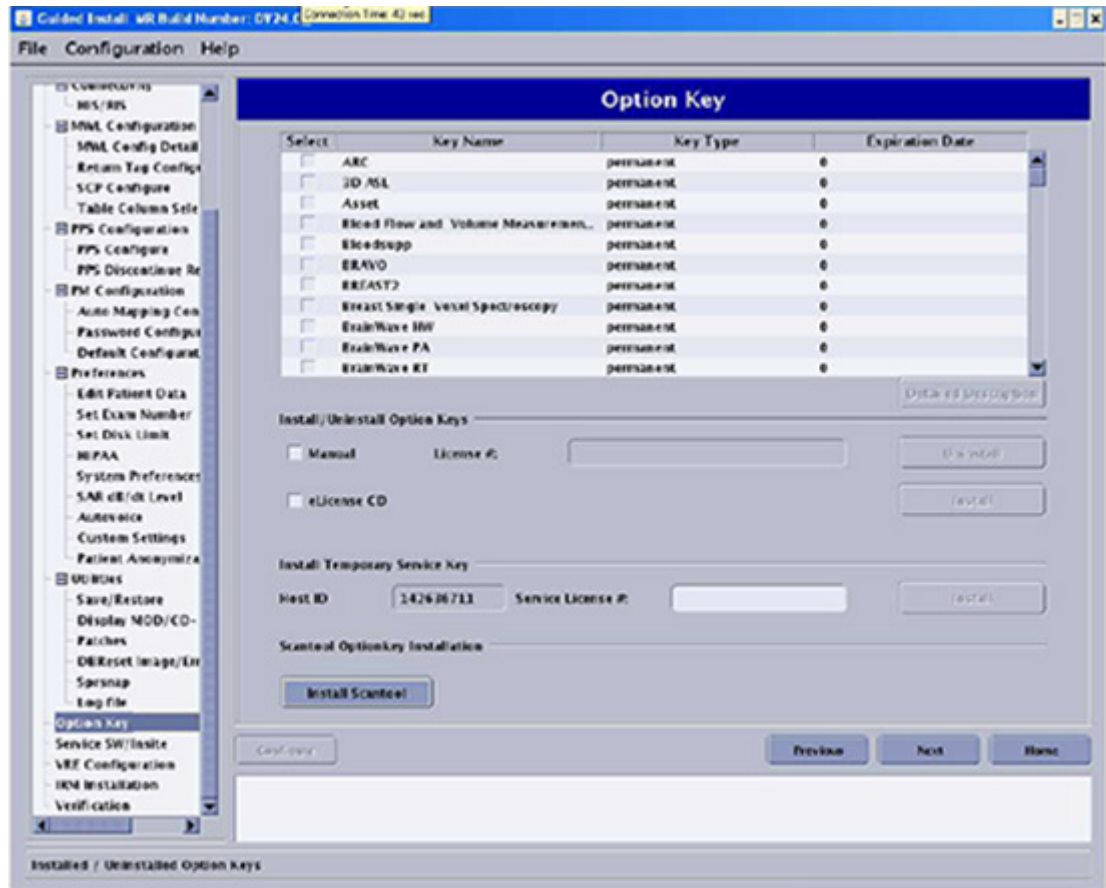
Illustration 2-7: Guided Install Hardware Configuration Menu



- When asked to load all applicable options, load the Silent MR option key. (This is not the Silent MRA option key)

- If upgrading from a pre-DV24 software version, select the “Install Scantool” button from the Option Key page of the “Guided Install” to get the full list of scan tools loaded onto the system.

Illustration 2-8: Guided Install Option Key page



3. Perform UPM Calibrations and UPM Functional Checks. Refer to Service Methods for procedure details. (Do not use the service key)
4. If upgrading from a pre-DV24 software version, insert service key and perform the HOEC Calibration. Refer to Service Methods for procedure details, otherwise skip this step.
5. If MR750w 3T GEM system, replace old GEM HNU coil with new 3.0T Quiet GEM HNU coil. Harvest the old HNU coil by using the packaging that the new Quiet GEM HNU coil came in for return shipping.
6. Run MCQA for any new coils.
7. For 1.5T/3T: Run a Quiet scan.
 - Select GE protocols tab
 - Filter list by: Quiet
 - Select adult_head_GEM Brain — Silent MR protocol
8. Perform SaveInfo — see service methods Save and Restore Tool

1.4 Completion

The following steps need to be completed by the GE Field Engineer.

1. Record and enter applicable data into applicable site configuration files and records.
2. Complete Product Locator information for all installed serialized components, new or updated, via either:

- (U.S. Only) FE Site Verification Web Site

or

- Process and return product locator installation cards for all serialized components to:

Product Locator File, P.O. Box 414, W-523, Milwaukee, WI 53201-0414

Refer to [Chapter 1, Product Locator](#) for details on submitting Product Locator information.

NOTE: Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

3. Make sure all replaced components are packaged and returned for recycling. Include the Test Procedure results.
4. Make sure all Service and Operator documentation is installed on the host computer.

Chapter 3 Appendix

1 Upgrade Test Record

Table 3-1: MR Quiet Upgrade Master Harvest Test Procedure

MR Quiet Upgrade					
System ID *		Dispatch Number *			
Cabinet Serial Number *		FE Name *			
		SSO *			
		Signature *			
		Test Date *			
Test Site Name *		Country *			
After test completion, send this document with harvested parts for review of test results. Completed test record should be signed and dated. Ensure all cells with * requiring information are completed or marked with "N/A" if not applicable.					
Manual Documentation Number		Revision Number	FE Confirm Rev.	If revision number is different, what rev. was used?	
5670014: Discovery MR450, MR750, Optima MR450w BASE & GEM, MR750w GEM Service Methods		2	*		
Test Number	Test & Procedure Documentation	Expected Results	Measured Results	Test Result Pass / Fail	Specific Failures
Test					
UPM Diagnostic Procedure Discovery MR450 & MR750, Optima MR450w BASE & GEM, MR750w GEM - <i>Troubleshooting->Diagnostics->System Function->RF->UPM Diagnostics</i>					
1	UPM Diagnostics I/O Data	Test completes	Test completes	*Circle One Pass / Fail	
2	UPM Diagnostics Analog Data	Test completes	Test completes	*Circle One Pass / Fail	
Patient Data Removal Verification					
3	Confirm Patient Data has been removed from the hard disc drives	N/A - System does not contain components which hold patient data		N/A	

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